

MARÍA MILAGROS LAHUERTA

Biomedical Engineering Master's Student

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Turin, Italy 15/08/2000 MARÍA MILAGROS LAHUERTA



PROFILE

Final-year Master of Science student in Biomedical Engineering, completing a double degree program between the National University of Córdoba and Politecnico di Torino. Seeking a Master's Thesis opportunity in neurotechnology and biomedical signal processing, with a strong interest in medical device development and translational research aimed at improving clinical practice.

EDUCATION

Double Master's Degree Politecnico di Torino
2025 – Present Turin, Italy

- Completing final M.Sc. year in Biomedical Engineering.
- Current Average: 26.6/30

Biomedical Engineering Degree National University of Córdoba
2020 – 2024 Argentina

- University Bachelor's Degree in Biomedical Engineering.

TECHNICAL SKILLS

- Biomedical Signal Processing
- Embedded Systems
- Microcontrollers
- Analog Electronics
- Amplifiers
- Filters
- Python
- MATLAB
- C
- C++

PROFESSIONAL SKILLS

- Analytical Thinking
- Research Mindset
- Independent Working
- Problem Solving
- Leadership
- Team Collaboration

LANGUAGES

Spanish: Native

English: Level B2 (Certified)

Italian: Level B2 (Certified)

German: Level B1 (Certified / Currently improving)

Mandarin Chinese: Basic

EXPERIENCE

Supervised Professional Internship Conci Carpinella Institute
May 2024 – Oct 2024 Córdoba, Argentina

- Acquired and preprocessed neurophysiological signals using a 128-channel EEG system, including subject preparation, electrode placement, impedance verification, signal quality assessment, and acquisition software.
- Participated in functional MRI (fMRI) studies, supporting experimental protocols and synchronization of task paradigms with MRI acquisition.

President (2025–2026) & VP (2024) IEEE UNC Student Branch
Feb 2024 – Feb 2026 Córdoba, Argentina

- Led multidisciplinary engineering teams, coordinating technical workshops, research initiatives, and collaborations with academia and industry.

Multidisciplinary Innovation Program iTeams – CONICET – Ministry of Science
Aug 2024 – Oct 2024 Córdoba, Argentina

- Evaluated technology transfer strategies to translate biomedical research into real-world healthcare applications.

Course “AI Tools in Medical Imaging” EMBS – IEEE / SABI
June 2024 Argentina

- Applied machine learning techniques for medical imaging analysis and computer-aided diagnosis.

Collaboration in In Vivo Experiments INICSA Institute
May 2024 – Oct 2024 Argentina

- Contributed to in vivo experimental protocols, biomedical data collection, and research on diabetes-related complications.

International Fellowship & Clinical Internship Friedrich Schiller Gymnasium & Pediatric Clinic
Sep 2016 – Jul 2017 Marbach am Neckar / Bietigheim-Bissingen, Germany

- Completed an international academic program, strengthening intercultural communication skills and German language proficiency.
- Gained first-hand exposure to the German healthcare system through a clinical internship in a pediatric hospital.

RELEVANT ENGINEERING PROJECTS

- Neural Signal Processing Using MEA Data:** Developed and validated a Python pipeline to process extracellular neuronal recordings from MEA datasets, implementing Butterworth filtering, spike/burst detection, feature extraction, and quantitative analysis to characterize network activity under developmental and pharmacological conditions.
- ECG Monitoring and Analysis Platform:** ECG Monitoring and Analysis Platform: Developed a Python telemedicine platform for visualization and analysis of multi-lead ECG recordings, applying digital signal processing techniques for R-peak detection, RR interval analysis, and heart-rate estimation from physiological signals.
- Embedded Brushless Motor Control System:** Developed an embedded control system in C using a PIC16F887 microcontroller, integrating a 4x4 matrix keypad, LCD interface, PWM motor speed control, and interrupt-driven RPM monitoring.
- Digital Twin for Remote Patient Monitoring:** Designed the architecture and user interfaces of a digital health platform integrating physiological sensor data, patient-reported outcomes, and clinical dashboards to support remote patient monitoring.



PORTFOLIO: milagroslahuerta.github.io
Additional projects, certificates and supporting material available online.